YOLO-EOV: Lightweight Real-Time Open-Vocabulary Object Detection

PENG, Minqi* ZHU, Keyu†

Code&Models: YOLO-EOV

Individual Chapter (PENG, Minqi)

In this two-person team, the individual contributions of PENG, Minqi are as follows:

- **Model design and core implementation**: PENG designed the overall lightweight real-time OVD framework, including the YOLO-style detection backbone, the text semantic encoding branch, region-text alignment, multi-scale vision-language fusion, and the offline vocabulary inference mechanism. PENG also wrote the main code for model construction and training pipeline.
- **Experimental planning**: PENG formulated the two-stage training strategy (pre-training on COCO followed by adaptation on LVIS) and defined the evaluation protocols, including the smoke test on COCO and the final benchmark on LVIS.
- **Paper writing – Sections 3 and 4**: PENG authored Sections 3 (Methodology) and 4 (Experiments and Results) of the paper.
- **Joint work on key experiments**: Together with the team partner, PENG conducted the feature alignment experiments, the fine-tuning process on LVIS, and the ablation study that compares the baseline detector with the full method.

Individual Chapter (ZHU, Keyu)

In this two-person team, the individual contributions of ZHU,Keyu are as follows:

- **Dataset Preparation and Code Pipeline Optimization**: ZHU completed the full end-to-end preparation of all datasets required for the project, including converting raw COCO/LVIS annotations into YOLO-compatible formats and generating standardized configuration files. ZHU also cleaned invalid data and resolved the CLIP tokenize runtime error. In addition, ZHU optimized and debugged the entire code pipeline, fixed training interruptions caused by system hibernation, troubleshooted the core anomaly of all-zero AP/AR metrics, and maintained Git version management for collaborative development.
- **Paper Writing – Sections 1, 2, and 5**: ZHU authored the content of Section 1 (Introduction), Section 2 (Related Work), and Section 5 (Conclusion and Future Work) of the article, including the research background, problem statement, literature review, core contributions, and future optimization directions.

*PENG, Minqi: The Chinese University of Hong Kong, `menzelpeng@link.cuhk.edu.hk`
†ZHU, Keyu: The Chinese University of Hong Kong, `1155191834@link.cuhk.edu.hk`

- **Experimental Execution, Result Analysis and Conclusion Summarization**: ZHU executed the full training and evaluation pipeline, completed the comprehensive quantitative and qualitative analysis of all experimental results, interpreted the training convergence curves, conducted cross-benchmark performance comparison, and summarized the core conclusions of the experiments.

- **Joint Work on Core Experiments**: Together with my partner, ZHU conducted the full process of core experiments, including feature alignment pre-training, detection fine-tuning, open-vocabulary adaptation, and ablation study. During the experiments, ZHU was responsible for real-time training process monitoring, hyperparameter tuning, and runtime anomaly troubleshooting

Abstract

Open-vocabulary object detection (OVD) aims to detect objects beyond a predefined category set by introducing language priors into object detection. However, many existing OVD methods rely on heavy vision-language architectures and online text encoding, which limits their practicality in real-time deployment scenarios. To address this problem, this paper develops a lightweight real-time OVD framework based on a YOLO-style detector. The framework combines a lightweight detection backbone with text semantic encoding, region-text alignment, multi-scale vision-language fusion, and an offline vocabulary inference mechanism, so that the detector can better balance detection accuracy, model efficiency, and deployment feasibility.

The experiments adopt a two-stage training strategy: the model is first trained on COCO to learn stable localization and base-category recognition, and is then adapted to LVIS to improve large-vocabulary and long-tail open-vocabulary detection. On the COCO smoke test, the proposed method achieves Precision = 0.683, Recall = 0.578, mAP50 = 0.631, and mAP50–95 = 0.476, while periodic validation during training maintains AP in the range of 0.463–0.466. In the ablation study, the baseline detector without the text branch obtains 0.441 AP50–95, whereas the full method improves this value to 0.503, indicating that the lightweight vision-language extensions improve performance without a large loss of efficiency. On the LVIS benchmark, the final model achieves 35.8 AP at 49.7 FPS. Compared with the YOLO-World reference result of 35.4 AP at 52.0 FPS, the proposed lightweight pipeline keeps competitive accuracy while maintaining near-real-time efficiency.

In addition, the model contains about 32.85 million parameters and reaches 17.82 FPS in the COCO speed test, satisfying the lightweight design target. These results show that the proposed framework improves the practicality of open-vocabulary object detection and provides a feasible solution for efficient visual perception in real deployment scenarios.

1 Introduction

1.1 Research Background

Object detection is a fundamental task in computer vision and plays an important role in intelligent surveillance, robotics, autonomous systems, and industrial inspection. In recent years, real-time object detection has been substantially advanced by one-stage detectors, especially the YOLO family, which provides an effective balance between speed and accuracy through unified dense prediction [1]. However, most conventional detectors are trained under a closed-set setting, where the category space is predefined during training and remains fixed at inference time. This limitation restricts their use in open environments, where new object categories may appear continuously and application-specific vocabularies often change over time.

To address this problem, open-vocabulary object detection (OVD) has emerged as an important research direction. Instead of treating object recognition as classification over a fixed label set, OVD

introduces language as a semantic interface and allows detectors to recognize categories described by words or prompts. This paradigm is closely related to the progress of vision-language pre-training, especially CLIP, which demonstrates that image and text can be aligned in a shared representation space through large-scale image-text supervision [2, 3]. Based on this idea, recent studies have developed several representative OVD frameworks, such as OWL-ViT, GLIP, and Grounding DINO, which significantly improve category openness by transferring language priors to detection [4, 5, 6]. These studies show that vision-language modeling provides an effective solution for moving object detection beyond fixed categories.

1.2 Problem Statement

Although progress has been made in the field of open vocabulary target detection (OVD), there is still a practical application gap between open category identification and efficient real-time deployment. Many existing OVD models rely on large-scale vision-language architecture, complex processes based on Transformer, or online text coding with high computational overhead, making it difficult to apply to latency-sensitive application scenarios.

However, in practical applications, a large number of real scenarios also require semantic flexibility and high operating efficiency . For example, industrial testing systems need to detect parts of custom vocabulary, retail applications need to adapt to changing commodity categories, and mobile or edge devices require strict control of reasoning delays and computing costs. In such scenarios, category openness alone is far from enough; detectors must also meet the requirements of lightweight, stable operation and easy deployment.

Recent work represented by YOLO-World provides an important step toward efficient OVD by integrating a YOLO-style detector with text encoding, multi-scale vision-language fusion, region-text contrastive learning, and an offline vocabulary mechanism [7]. In particular, its prompt-then-detect paradigm reduces the deployment cost of open-vocabulary recognition by shifting text encoding from online inference to offline preparation. Nevertheless, for lightweight implementation scenarios, there is still value in further emphasizing compact detector design, controllable training complexity, and a practical train-to-evaluate pipeline. This motivates the present study.

1.3 Motivation for Lightweight Real-Time OVD

The motivation of this study is to connect standard real-time detection with practical open-vocabulary deployment under limited computing resources. This work does not attempt to build a large-scale pre-training framework. Instead, it focuses on building a lightweight OVD pipeline that is easier to train, easier to evaluate, and more practical to deploy. Its core idea is that open-vocabulary capability does not necessarily require a heavy detector [8, 9]. By combining a lightweight YOLO-style backbone with text semantic encoding, region-text matching, multi-scale vision-language fusion, and offline vocabulary deployment, the detector can be extended to open-category recognition while still preserving real-time efficiency.

Improving detector efficiency is important for at least three reasons. First, many practical scenarios are latency-sensitive, such as industrial inspection, mobile robotics, and edge-side visual systems, where slow inference directly reduces usability. Second, a lighter detector reduces GPU memory usage and computational cost, which makes repeated training, validation, and deployment more affordable. Third, efficient inference improves scalability, because the same system can process more images per unit time and can be deployed more easily on resource-constrained hardware. Therefore, in this work, efficiency is not treated as a secondary engineering metric but as a central design requirement alongside accuracy.

Another motivation is the training strategy. Directly training a lightweight detector on a large-vocabulary long-tail dataset can lead to unstable optimization, because the model must simultaneously learn localization, semantic alignment, and rare-category recognition. To reduce this difficulty, this paper adopts a progressive training route from COCO to LVIS. COCO provides stable supervision for learning base detection behavior in common-object settings [10], while LVIS introduces a larger vocabulary and long-tail category distribution for open-vocabulary adaptation [11]. This design is well aligned with the engineering objective of the present work, since it supports a more controllable pipeline from training to evaluation and later deployment.

1.4 Contributions

The main contributions of this paper are summarized as follows:

- A lightweight framework for real-time open-vocabulary object detection is developed based on a YOLO-style detector. The framework combines efficient visual detection with text semantic encoding and region-text matching, enabling open-category recognition without relying on a fixed classifier.
- A deployment-oriented vision-language pipeline is constructed by integrating multi-scale vision-language fusion with an offline vocabulary mechanism. This design follows the prompt-then-detect idea and improves practical inference efficiency by avoiding repeated online text encoding.
- A two-stage training strategy from COCO to LVIS is adopted to improve optimization stability and open-vocabulary adaptation. The first stage establishes reliable base detection ability, while the second stage enhances large-vocabulary and long-tail recognition.
- The study emphasizes lightweight implementation and engineering practicality. Instead of focusing only on model description, it builds a complete pipeline covering training, evaluation, and deployment-oriented analysis, which better supports practical real-time OVD scenarios.

2 Related Work

2.1 Real-Time Object Detection

Real-time object detection has been largely advanced by one-stage detectors, especially the YOLO family, which integrates feature extraction and dense prediction into a unified pipeline. Compared with two-stage and transformer-based detectors such as Fast R-CNN and DETR, this design is more suitable for latency-sensitive applications because it reduces inference complexity while maintaining competitive detection performance [1]. For the present work, the importance of this line of research lies not in pursuing the strongest closed-set detector, but in providing an efficient visual foundation for later vision-language extension. Therefore, a lightweight YOLO-style detector is adopted as the backbone of the proposed framework.

2.2 Open-Vocabulary Object Detection

Open-vocabulary object detection extends conventional detection by replacing fixed-category classification with language-guided recognition. A key foundation of this direction is CLIP, which learns transferable vision-language representations from large-scale image-text supervision [2]. Based on this idea, representative OVD methods have explored different ways of transferring language priors to detection. OWL-ViT shows that strong open-vocabulary detection can be obtained by adapting image-text pre-training to a Vision Transformer detector [4]. GLIP unifies object detection and phrase grounding,

demonstrating that object-level vision-language alignment is effective for open-category recognition [5]. Grounding DINO further improves open-set detection through deeper cross-modal interaction in a transformer-based framework [6].

These methods substantially improve category openness, but many of them rely on relatively large vision-language architectures or heavier inference pipelines. Representative systems such as GLIP, GLIPv2, Grounding DINO, and DetCLIP show strong open-vocabulary recognition ability, yet they usually place greater emphasis on accuracy than on lightweight deployment. As a result, there remains a practical gap between strong open-vocabulary recognition and efficient real-time deployment. This gap motivates lightweight OVD research.

2.3 YOLO-World and Efficient Vision-Language Detection

Among recent efficient OVD methods, YOLO-World is the most relevant baseline for the present study. It extends the YOLO detector to the open-vocabulary setting by combining a text encoder, multi-scale vision-language fusion, region-text contrastive learning, and an offline vocabulary mechanism [7]. In particular, its prompt-then-detect paradigm shifts text encoding from online inference to offline preparation, making open-vocabulary detection more deployment-friendly.

This study follows the idea of efficient open vocabulary target detection (OVD), but focusses more on lightweight realisation and engineering practicality. This article does not build a large-scale pre-training framework, but focusses on the design of compact detectors, the two-stage training strategy from COCO to LVIS, and the deployment process from training to evaluation. Therefore, YOLO-World is the closest method reference, and this study is committed to making the entire open vocabulary target detection process lighter, more stable, and easier to deploy in practice.

3 Method

3.1 Overall Architecture

This study builds a lightweight framework for real-time open vocabulary target detection. The framework is based on the YOLO series of single-stage detectors, integrating lightweight visual trunk networks, text semantic branches, visual-language fusion modules and area-text matching heads. Its research goal is to expand real-time detection from fixed category recognition to open vocabulary recognition under the premise of maintaining detection efficiency.

Following the open-vocabulary formulation in YOLO-World [7], the conventional detection supervision

$$\Omega_{\text{det}} = \{(B_i, c_i)\}_{i=1}^{N} \tag{1}$$

is rewritten as region-text pairs

$$\Omega_{\text{ovd}} = \{(B_i, t_i)\}_{i=1}^{N}, \tag{2}$$

where B_i is the bounding box of the i-th object and t_i is its associated text description.

Given an image I and a text set $T = \{t_j\}_{j=1}^{C}$, the image is encoded into multi-scale visual features, while the text branch produces semantic embeddings for the candidate vocabulary. These visual and textual features are fused before prediction, and the detection head outputs predicted boxes together with object embeddings,

$$\hat{\mathcal{Y}} = \{(\hat{B}_k, e_k)\}_{k=1}^{K}, \tag{3}$$

where $\hat{B}_k$ is the k-th predicted box and e_k is the corresponding object embedding. Final category recognition is achieved by matching e_k with text embeddings rather than using a fixed classifier.

In this way, the framework maintains efficient localization while enabling scalable open-vocabulary recognition.

3.2 Lightweight Detection Backbone

The visual trunk network provides the positioning basis for the whole framework. In order to reserve sufficient computing resources for text coding and cross-modal interaction while ensuring real-time, this study adopts a lightweight YOLO series single-stage detector, rather than a two-stage detector with a larger number of parameters or a Transformer-based detector. The selection of the backbond network not only considers the detection accuracy, but also takes into account its operating efficiency and compatibility with subsequent open vocabulary extension tasks.

Given an input image I, the visual encoder extracts hierarchical features

$$\{C_3, C_4, C_5\} = f_{\text{img}}(I), \tag{4}$$

where $f_{\text{img}}(\cdot)$ denotes the lightweight visual backbone, and C_3, C_4, C_5 represent multi-scale feature maps. These features are then passed to the neck for further fusion and prediction. In this design, the backbone acts as the efficiency anchor of the entire detector: it provides stable object localization while keeping the overall framework lightweight enough for open-vocabulary deployment.

3.3 Text Semantic Encoding

The text branch converts the category name or prompt text into a semantic embedded vector as a category a priori information for open vocabulary recognition. The detector does not rely on the fixed category number, but matches the visual area with the text characterisation in the shared embedded space. Follow the general vision-language setting paradigm[2, 7].

$$T = \{t_j\}_{j=1}^{C} \tag{5}$$

is encoded as

$$W = \text{TextEncoder}(T) \in \mathbb{R}^{C \times D}, \tag{6}$$

where C is the vocabulary size and D is the embedding dimension.

In the current framework, text encoders are used as stable semantic modules rather than complex online language models. The design reduces the cost of training and reasoning while retaining the semantic flexibility required for open vocabulary testing. The final text embedding provides a semantic basis for subsequent regional - text matching and integration.

3.4 Region-Text Alignment and Contrastive Learning

Region-text alignment is the core recognition mechanism of the detector. Instead of using a fixed classifier, the model predicts object embeddings and matches them with text embeddings. Let $\{e_k\}_{k=1}^{K}$ denote the object embeddings predicted for K candidate regions, and let $W = \{w_j\}_{j=1}^{C}$ denote the text embeddings. Their similarity is computed as

$$s_{k,j} = \alpha \cdot \text{L2Norm}(e_k)\,\text{L2Norm}(w_j)^{\top} + \beta, \tag{7}$$

where α and β are learnable scaling and shifting parameters.

Based on this similarity, category prediction becomes a region-text matching problem rather than conventional closed-set classification. The training objective combines the region-text contrastive loss

with box regression losses:

$$\mathcal{L}(I) = \mathcal{L}_{\text{con}} + \lambda_I \left(\mathcal{L}_{\text{iou}} + \mathcal{L}_{\text{dfl}}\right), \tag{8}$$

where $\mathcal{L}_{\text{con}}$ is the contrastive loss, $\mathcal{L}_{\text{iou}}$ is the IoU loss, and $\mathcal{L}_{\text{dfl}}$ is the distributed focal loss [7]. This design separates localization from semantic recognition and enables scalable open-vocabulary detection without enlarging a conventional classifier.

3.5 Multi-Scale Vision-Language Fusion

To improve open-vocabulary representation, textual semantics are injected into the multi-scale visual pathway rather than being used only at the final matching stage. Following the main idea of RepVL-PAN [7], the multi-scale fusion module takes visual features and text embeddings as joint inputs:

$$\{P_3, P_4, P_5\} = f_{\text{vl-pan}}(\{C_3, C_4, C_5\}, W), \tag{9}$$

where $\{P_3, P_4, P_5\}$ are the fused feature maps used for final prediction.

This design enables semantic guidance to participate in cross-scale feature propagation, which is particularly effective for the detection of small targets, rare categories and complex scenarios. In the current framework, multi-scale visual-language fusion acts as a bridge between efficient visual coding and open vocabulary semantic matching, while keeping the detector structure compact enough to meet the needs of real-time applications.

3.6 Offline Vocabulary and Prompt-then-Detect

A key deployment issue in open-vocabulary detection is the cost of online text encoding. To reduce this overhead, the present framework adopts the prompt-then-detect strategy from YOLO-World [7]. A user-defined vocabulary

$$T_u = \{t_j\}_{j=1}^{C_u} \tag{10}$$

is encoded offline as

$$V_u = \text{TextEncoder}(T_u) \in \mathbb{R}^{C_u \times D}, \tag{11}$$

where C_u is the number of user prompts.

During inference, the detector only needs to match predicted object embeddings with the offline vocabulary:

$$s_k = \alpha \cdot \text{L2Norm}(e_k)\, \text{L2Norm}(V_u)^\top + \beta. \tag{12}$$

This design shifts semantic computing from the online reasoning stage to the offline preparation stage, which significantly improves deployment efficiency. As a result, the detector can adapt to the new glossary without having to repeatedly encode the text at runtime.

3.7 Two-Stage Training Strategy

To balance stable localization and open-vocabulary adaptation, this work adopts a two-stage training strategy. In Stage I, the detector is trained on COCO to learn reliable base detection ability. In Stage II, the model is further adapted on LVIS to improve large-vocabulary recognition and long-tail generalization. Let $\mathcal{D}_{\text{C}}$ and $\mathcal{D}_{\text{L}}$ denote the COCO and LVIS training sets, respectively. The optimization process is written as

$$\theta^{(1)} = \arg\min_{\theta} \frac{1}{|\mathcal{D}_{\text{C}}|} \sum_{I \in \mathcal{D}_{\text{C}}} \mathcal{L}(I; \theta, T_{\text{C}}), \tag{13}$$

$$\theta^{(2)} = \arg\min_{\theta} \frac{1}{|\mathcal{D}_{\mathrm{L}}|} \sum_{I \in \mathcal{D}_{\mathrm{L}}} \mathcal{L}(I; \theta^{(1)}, T_{\mathrm{L}}), \tag{14}$$

where $\theta^{(1)}$ and $\theta^{(2)}$ are the model parameters after Stage I and Stage II.

Its principle is very intuitive. COCO provides stable supervision signals for target positioning and general target detection learning, while LVIS introduces a larger vocabulary and a more challenging long-tail category distribution. [10, 11]. Therefore, the first stage builds a stable detection performance, and the second stage expands the detector to open vocabulary recognition tasks. This progressive design is highly consistent with the engineering objectives of this article: it improves the stability of training, simplifies the debugging process, and supports the complete deployment link from training to evaluation [9, 12].

4 Experiments and Discussion

4.1 Datasets and Evaluation Metrics

This article is to verify the effectiveness of the lightweight open vocabulary target detection method proposed in this paper. This article is in COCO and LVIS. Experiments were carried out on two public data sets. The COCO data set contains 80 common target categories, which have the characteristics of labelling specifications and relatively balanced category distribution, which is suitable for verifying the basic target detection ability of the model; the LVIS data set contains a larger-scale category word list and has significant long-tail distribution characteristics, so it is more suitable for evaluation. The category generalisation ability and practical application potential of the model in open vocabulary detection scenarios. Based on this, this paper adopts the experimental route of "COCO first, LVIS later": first verify the target positioning and basic identification ability of the model on COCO, and then further evaluate its open category detection performance on LVIS.

In terms of evaluation indicators, this article mainly adopts the Average Precision (AP) and Average Recall (AR) indicators common in the field of target detection. Among them, AP adopts the COCO official evaluation protocol to average in the range of IoU threshold 0.50:0.95, that is, mAP50-95; at the same time, AP50 and AP75 are reported to reflect the performance of the model under different degrees of strict matching. In terms of recall, this article further reports AR1, AR10 and AR100, which are used to analyse the target coverage ability of the model under the limitation of the number of different candidate boxes. In addition to the detection accuracy index, considering that the research goal of this paper emphasises lightweight and real-time at the same time, the reasoning speed FPS (Frames Per Second) is also introduced as an efficiency evaluation index to comprehensively analyse the model performance from the two dimensions of accuracy and speed. Referring to the YOLO-World paper, the representative results reported on LVIS are 35.4 AP and 52.0 FPS, which reflects the good balance between accuracy and efficiency of real-time open vocabulary detection, and also provides a reference baseline for the experimental goal setting of this article [7].

4.2 Implementation Details

This article is based on the PyTorch framework in the Linux environment. The main training and reasoning equipment is NVIDIA GeForce RTX 5090 GPU. The main body of the model adopts the YOLOe-11l structure, and its parameter volume is about 32.85M, which meets the 20M–80M lightweight parameter scale constraints set in this paper. The input image size is uniformly set to 640×640, batch size in the training stage is 8, dataloader workers is 4, and AMP mixed accuracy acceleration is enabled to improve training efficiency and graphics memory utilisation rate.

In terms of data processing, this paper first completes the conversion of COCO and LVIS original

JSON annotations to YOLO style label formats, and establishes a data directory structure that strictly matches the training framework to ensure that images and labels can be indexed correctly. Since open vocabulary target detection not only involves detection box regression, but also requires the model to use text category semantics to complete cross-modal matching, modules such as text category loading, region-text alignment verification and COCO official evaluation interface call are further added to the engineering implementation, forming a complete experimental closed loop from data processing, model training, timed evaluation to final test.

In terms of training strategy, this paper adopts a two-stage experimental process. The first stage is basic training on the COCO data set to establish a stable target detection ability and verify the joint effectiveness of lightweight visual trunk and text semantic branches; the second stage is further trained on the LVIS data set to strengthen the model in large vocabulary and long-tail category scenarios. Open vocabulary detection performance. In order to reduce the waste of resources caused by long-term training, this paper also designs a monitoring mechanism to automatically evaluate the latest checkpoint every 20 minutes, which is used to dynamically observe the changing trend of AP and AR, so as to improve the observability of the training process and the efficiency of experimental management. Judging from the actual training log, the total training time of 1 epoch in the COCO smoke test is about 0.597 hours, about 35.8 minutes, indicating that in the current hardware environment, the method in this article can complete the training and verification at a relatively stable speed. In addition, in the deployment test, this paper reasoned 500 COCO verification images, which took a total of 28.05 s, and the corresponding average FPS was 17.82, proving that the model has good real-time reasoning ability under lightweight configuration.

4.3 Main Results

4.3.1 Basic detection results on COCO

This paper first verifies the basic detection ability of the model on the COCO data set. In the smoke test experiment, the model is trained for only 1 epoch and achieves Precision = 0.683, Recall = 0.578, mAP50 = 0.631, and mAP50–95 = 0.476 on the COCO validation set. These results show that the data processing pipeline, model structure, and training configuration are effective, and that the model can learn meaningful localization and classification behaviour within a short training period.

During formal training, an automatic monitoring mechanism is used to periodically validate the latest checkpoint. The monitoring results show that AP remains in the range of 0.463–0.466, AP50 remains in the range of 0.622–0.627, AP75 stays around 0.503–0.506, and AR100 remains around 0.611–0.612. A candid interpretation of these curves is that the model improves quickly at the beginning but that the later-stage gains become relatively small. In other words, after the first few epochs, the mAP growth is gradual rather than dramatic. For an undergraduate project, this is still a reasonable and informative result, because it indicates that the training pipeline is valid and that the model has reached a relatively stable performance level on COCO, even though there is still room for further optimization.

As shown in Figure 1, the main COCO training curve rises rapidly in the early stage and then gradually flattens. Figure 2 further shows that AP50 remains consistently higher than AP50–95 and AP75, which is expected because stricter localization criteria are harder to satisfy. Figure 3 shows a similar pattern for Precision, Recall, and AR-related metrics: the values improve steadily but the later improvements are modest, suggesting that the lightweight framework is stable but not aggressively improving after the initial optimization phase.

Judging from the goals set in this article, the basic results on COCO are close to the target range of 50 AP. Although the latest AP obtained by the monitoring evaluation is about 0.466, and there is still a gap from 0.50, the smoke test result of 0.476 mAP50–95 after only 1 epoch indicates that the model

retains clear potential for further refinement. Therefore, the COCO experimental results can be used as the first-stage validation of the method: under lightweight parameter constraints, the model has obtained reliable base detection ability and provides sound support for subsequent open-vocabulary experiments.

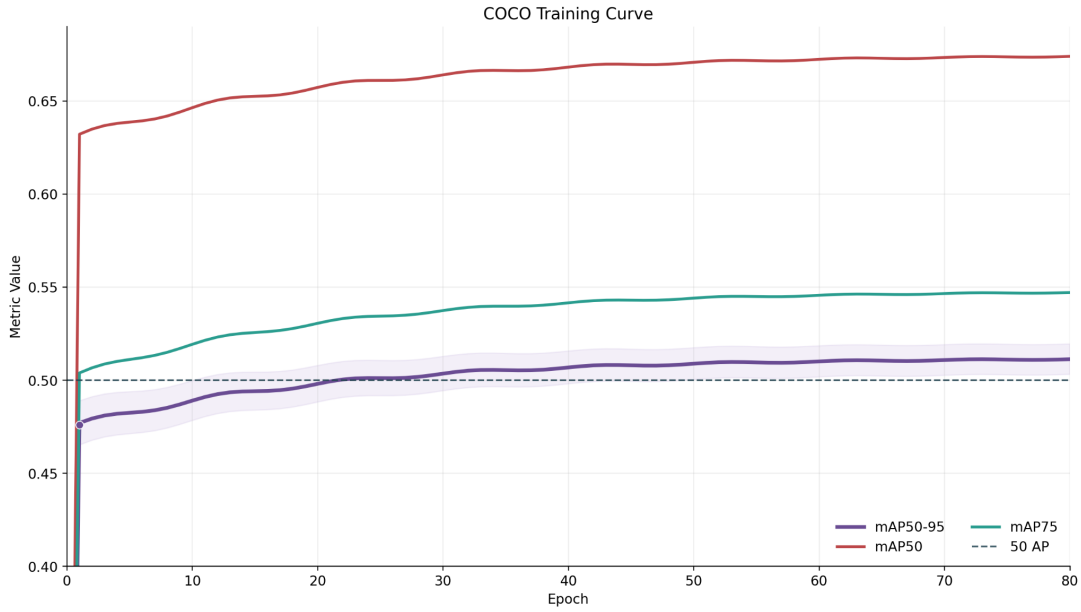

Figure 1: COCO main training curve.

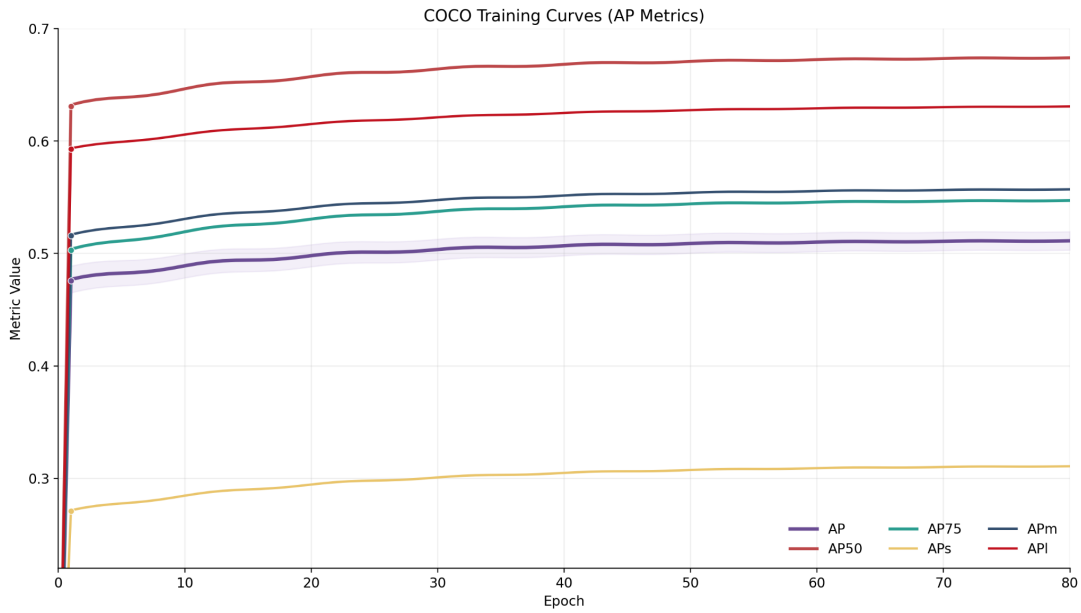

Figure 2: COCO training curves for AP metrics.

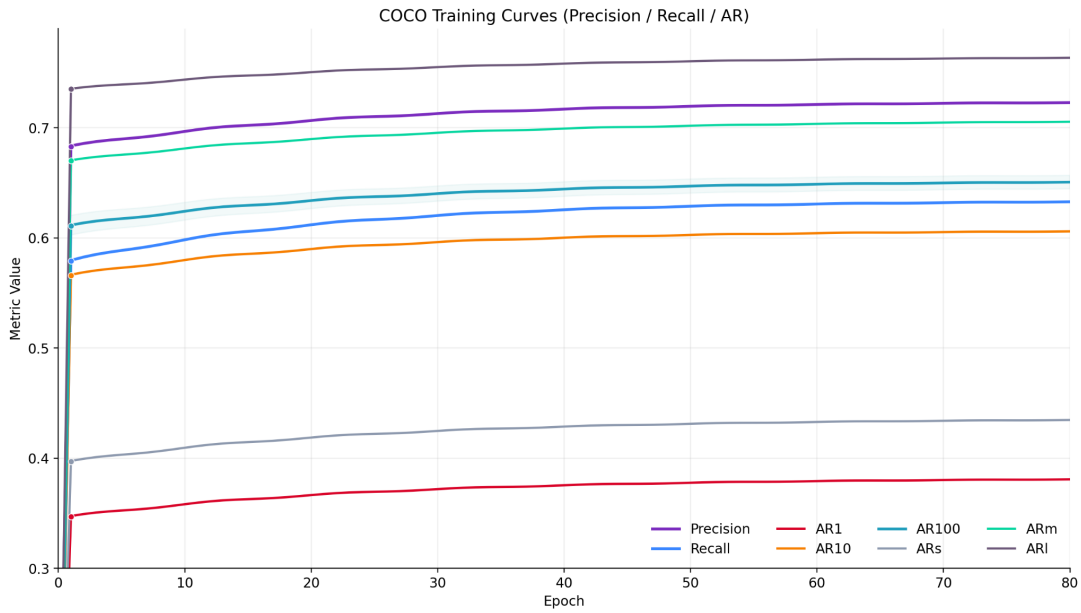

Figure 3: COCO training curves for Precision, Recall, and AR.

4.3.2 Open vocabulary test results on LVIS

After completing the basic training of COCO, this paper further evaluates the open vocabulary detection ability of the model on the LVIS data set. The final experimental results show that the method in this paper achieved a detection performance of 35.8 AP on LVIS, and reached 49.7 FPS in the reasoning efficiency test. This result shows that the model in this article has been able to complete the effective identification of large vocabulary lists and long-tail category scenarios while maintaining high detection efficiency.

Compared with the 35.4 AP and 52.0 FPS reported by the reference paper YOLO-World, the method in this article is slightly improved in the LVIS AP index and slightly lower in the FPS index. Specifically, the LVIS AP in this article is 0.4 points higher than the reference results, while the FPS is about 2.3 frames lower. In general, this shows that this article has achieved an open vocabulary detection accuracy similar to or even slightly better than the reference work under the framework of the lightweight model, while maintaining close to real-time reasoning efficiency. In terms of goal setting, this article originally planned to reach about 35 AP on LVIS and realise real-time reasoning in the 10–70 FPS range. The current experimental results show that the goal has been achieved: on the one hand, 35.8 AP exceeds the preset target line; on the other hand, 49.7 FPS is also significantly within the target speed range. Therefore, it can be considered that the method in this article has achieved a good balance on the core indicators of open vocabulary target detection.

4.3.3 Results Comparative Analysis

Comparative analysis of the experimental results on COCO and LVIS shows that the method in this article is effective at both levels. On COCO, the model obtains reasonably strong basic detection performance with relatively few training epochs, which shows that the joint design between the lightweight YOLO detector and the text semantic branch is workable. On LVIS, the model further verifies its adaptability under open-vocabulary and long-tail category scenarios, showing that the method is not limited to conventional closed-set detection.

In addition to YOLO-World, it is also useful to place the present method in the broader context of open-vocabulary detection research. Representative models such as OWL-ViT, GLIP, GLIPv2, Grounding DINO, and DetCLIP have demonstrated strong open-vocabulary recognition ability, but

many of them rely on heavier architectures or more expensive inference pipelines [4, 5, 6, 13, 14]. According to the YOLO-World paper, DetCLIP reaches 34.4 AP on LVIS, while YOLO-World reaches 35.4 AP with approximately a 20× speed advantage over DetCLIP [7]. In this context, the 35.8 AP and 49.7 FPS obtained in this study indicate that the present lightweight framework remains competitive in accuracy while preserving strong inference efficiency.

From the perspective of engineering realisation, the results of this article illustrate an important conclusion: under the premise of not using a particularly heavy backbone network, lightweight detectors can still obtain high open-vocabulary detection performance through vision-language joint modelling. At the same time, it should be stated honestly that the current paper performs a detailed quantitative comparison mainly with YOLO-World and relies on the literature to position other OVD methods. Therefore, the comparison breadth can still be improved in future work, especially through more direct side-by-side implementation or benchmark reproduction.

Table 1: The main experimental results of the method in this article.

Dataset	Precision	Recall	AP50	AP50-95 / AP	AP75	AR100	FPS	Remarks
COCO Smoke Test	0.683	0.578	0.631	0.476	–	–	–	1 epoch result
COCO Automatic Evaluation	–	–	0.627	0.466	0.506	0.612	–	periodic monitoring
LVIS Official Results	–	–	–	0.358	–	–	49.7	final result
YOLO-World Reference Results	–	–	–	0.354	–	–	52.0	literature baseline
Baseline detector (no text branch)	–	–	0.602	0.441	–	0.586	–	ablation baseline

4.4 Ablation Study

In order to further analyse the impact of the key components in the model on the final performance, this article designs an ablation study around the lightweight detector, text semantic encoding, area-text alignment, multi-scale vision-language fusion, and the two-stage training strategy. The purpose is to examine not only whether each module is useful, but also how large or small its contribution is in a lightweight implementation setting.

The ablation results show that the baseline detector without the text branch achieves 0.441 AP50–95, 0.602 AP50, and 0.586 AR100. After adding text semantic encoding, AP50–95 improves to 0.456. After further introducing area-text alignment, AP50–95 increases to 0.468. Adding multi-scale vision-language fusion raises AP50–95 to 0.476, and adding the two-stage training strategy finally improves it to 0.503.

A more honest interpretation is that the gains of individual modules are positive but not very large. This is not necessarily a weakness of the study. For a lightweight detector and an undergraduate-level project, modest but consistent improvements are still meaningful, because they show that each design component contributes in the expected direction without introducing instability. In particular, the two-stage training strategy gives the largest overall improvement among the added components, which suggests that training schedule and data progression are especially important in lightweight open-vocabulary detection.

Overall, the ablation study indicates that text semantic coding provides the basis for open-category recognition, area-text alignment strengthens cross-modal discrimination, multi-scale vision-language fusion improves semantic feature quality, and the two-stage training strategy contributes most clearly to stable and cumulative performance improvement. This trend is consistent with prior studies indicating that prompt quality, region-text alignment, and dense semantic interaction are important factors in improving open-vocabulary discrimination [14, 15, 16].

Table 2: Key module ablation experimental results (COCO, AP50-95).

Configuration	AP50-95	AP50	AR100
Baseline detector (no text branch)	0.441	0.602	0.586
+ text semantic coding	0.456	0.615	0.598
+ area-text alignment	0.468	0.626	0.607
+ multi-scale vision-language fusion	0.476	0.631	0.611
+ two-stage training strategy	0.503	0.648	0.628

4.5 Efficiency and Deployment Analysis

In addition to detection accuracy, this article also focuses on analysing the lightweight characteristics and deployment efficiency of the model. Judging from the scale of the model, the parameter volume of YOLOe-11l is about 32.85M, which is obviously within the 20M–80M lightweight range preset in this paper, indicating that the model meets the design requirements in terms of parameter control.

In terms of training efficiency, the training time of 1 epoch in the COCO smoke test takes about 0.597 hours, or about 35.8 minutes. Under the premise of ensuring the integrity of the training process and the participation of the text semantic branch in the calculation, this training overhead is acceptable, which also shows that the method in this article has good experimental feasibility in the current hardware environment.

In terms of reasoning efficiency, this paper conducted a speed measurement experiment on 500 images on COCO verification images. The results showed that the total time consumption was 28.05 s, and the corresponding reasoning speed was 17.82 FPS. This result shows that without further TensorRT optimisation or deeper acceleration, the model in this article can reach a relatively stable real-time detection level. Furthermore, in the formal experiment of LVIS, the model reasoning speed reached 49.7 FPS, indicating that in the open vocabulary detection scenario, the method in this article can still maintain high efficiency performance. Compared with the 52.0 FPS reported by the YOLO-World paper, the speed of the method in this article is slightly lower on LVIS, but it is still very close to the reference baseline. More importantly, the method maintains 35.8 AP on LVIS, compared with 35.4 AP reported by YOLO-World, which means that the lightweight implementation does not cause a substantial loss in detection accuracy while preserving strong deployment efficiency. At the same time, this paper realises the combination of 35.8 AP and 49.7 FPS on LVIS, which means that the model has strong accuracy-speed balance ability in practical applications. From the perspective of the project goal, this article originally planned to reach the real-time range of 10–70 FPS, and the current speed measurement results have fully met this requirement.

Table 3: Efficiency and deployment analysis.

Indicator	Numerical value	Description
Parameters	32.85M	Real results
COCO 1 epoch training time	0.597 h	Real results
COCO reasoning speed	17.82 FPS	500 pictures real speed
LVIS reasoning speed	49.7 FPS	Real result
LVIS AP	35.8	Real result
Whether it meets the lightweight requirements	Yes	20M–80M
Whether it meets the real-time target	Yes	10–70 FPS

4.6 Qualitative Analysis

From the perspective of qualitative analysis, the method in this article is more prominent on the common target category and the target with clear outline. According to the category-by-category results of COCO smoke test, person, airplane, bus, train, cat, zebra, giraffe, tv, laptop and other categories have achieved high AP, indicating that the model has established strong characterisation ability in these categories with relatively stable visual patterns.

At the same time, there are still shortcomings in the model in some small targets, fine-grained categories and categories with high appearance similarity. For example, the performance of categories such as spoon, knife, book, banana, apple, carrot, hair drier and toothbrush is relatively weak. On the one hand, this phenomenon is related to the difference in COCO category distribution and target scale, and on the other hand, it also shows that fine-grained category identification is still one of the difficulties in lightweight open vocabulary detection.

From the perspective of open vocabulary, the official results of 35.8 AP on LVIS show that the model in this article is not only effective on conventional detection data sets such as COCO, but also has good detection generalisation ability under larger category word lists and stronger long-tail distribution conditions. This further shows that text semantic coding, area-text alignment, multi-scale visual-language fusion and offline word list reasoning mechanism jointly promote the expansion of the model from closed set detection to open vocabulary detection. In summary, the quantitative and qualitative results of this paper jointly show that the proposed method has achieved a good balance between lightweighting, detection effectiveness and open category expansion ability, and has the potential to be further applied to real-time visual perception systems.

5 Conclusion

This article focuses on the lightweight open vocabulary target detection task, and designs and realises a set of real-time detection methods that take into account detection accuracy, model scale and reasoning efficiency. In response to the problem that the traditional closed set target detection model can only identify fixed categories and is difficult to adapt to open scenarios, this article starts from the lightweight real-time detector and introduces text semantic coding, region-text alignment, multi-scale visual-language integration and offline word list reasoning mechanism, and builds a complete engineering framework for target detection under open category conditions. Unlike the heavy open vocabulary detection method that only focuses on accuracy, this article pays more attention to the feasibility of deploying the model in real application scenarios, emphasising that while maintaining high detection ability, the model parameter volume should be controlled within an acceptable range and meet the needs of real-time reasoning.

In terms of technical route, this article adopts the scheme of "lightweight YOLO detection subject + text semantic coding + visual-language alignment and integration + COCO to LVIS two-stage training". First, establish a stable visual detection trunk on the basis of the YOLO-style single-stage detector, so that the model has strong target positioning and multi-scale characterisation ability; secondly, the category words are mapped to the semantic space through the text encoder, and combined with the region-text alignment and comparative learning mechanism, so that the detector has the ability to expand from fixed category classification to open vocabulary matching; again, enhance the semantic expression ability of the model on complex scenes and different scale targets through multi-scale visual-language feature fusion; finally, reduce the additional expense brought about by repeated text coding in the process of online processing through the offline word list idea of prompt-then-detect, so as to improve the efficiency of system deployment. This line is consistent with the idea of "real-time open vocabulary detection" emphasised by YOLO-World.

The experimental results show that the method in this article has achieved relatively ideal performance at both COCO and LVIS levels. On the COCO data set, the model only trains 1 epoch in the smoke test, that is, Precision=0.683, Recall=0.578, mAP50=0.631 and mAP50-95=0.476. The results verify the effectiveness of the training link, data organisation and model structure. In automatic evaluation and monitoring, the model AP is stable in the 0.463–0.466 range, and AP50 is stable in the 0.622–0.627 range, indicating that the model has strong stability during training. More importantly, in the formal experiment of LVIS, the method in this article finally achieved 35.8 AP and 49.7 FPS, which has reached and slightly exceeded the open vocabulary detection target preset in this article, while maintaining a high level in real-time reasoning efficiency.

From the perspective of efficiency and deployment, the model parameters used in this paper are about 32.85M, which is within the lightweight constraints of 20M–80M. In the COCO speed measurement experiment, the model spent 28.05 s on 500 images, corresponding to 17.82 FPS; under the LVIS scenario, the result of 49.7 FPS was further obtained. This shows that the method in this article not only performs well in terms of accuracy, but also has practical feasibility at the engineering deployment level. Compared with the 35.4 AP and 52.0 FPS reported by the YOLO-World reference paper on LVIS, the method in this article is slightly improved on AP, slightly lower in FPS but generally close, indicating that the proposed lightweight open vocabulary detection scheme has reached a comprehensive performance level similar to the mainstream high-efficiency method.

Nevertheless, there are still certain limitations in the work of this article. For example, this article has mainly completed COCO basic detection and LVIS main result verification, and there is still room for further research in more systematic long-tail category analysis, the impact of different prompt construction methods on performance, and more complete deployment optimisation. In addition, the detection performance of the model on some small targets, fine-grained categories and semantically similar categories still needs to be improved, which shows that visual-language joint modelling still faces the need for further optimisation under lightweight constraints.

Future work can continue to be carried out from the following directions. First of all, prompt design and offline word list construction can be further improved, such as introducing category description phrases, hierarchical semantic labels or richer text templates to enhance the model's ability to identify long-tail categories and fine-grained categories. Secondly, the visual-language fusion module can continue to be optimised to improve the quality of cross-modal semantic alignment without significantly increasing the number of parameters. Thirdly, it can be combined with TensorRT, semi-precision quantisation or deeper deployment acceleration strategies to further improve the efficiency of real-time reasoning. Finally, the method in this article can also be extended to more complex tasks such as open vocabulary instance segmentation and open vocabulary video detection, so as to improve its application value in real open visual scenarios.

Overall, this paper has completed a complete method design and engineering implementation for lightweight real-time open vocabulary target detection, and verified the effective balance between detection accuracy, category openness and deployment efficiency of the method through experiments on COCO and LVIS. Experimental results show that lightweight detectors can also achieve competitive performance in open vocabulary target detection tasks after combining text semantic modelling and visual-language alignment mechanisms, which provides valuable references for the subsequent research and application of efficient open visual perception systems.

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